

REBRIKHA, Russian Federation

WMO: 299230

Lat: 53.082N

Lon: 82.338E

Elev: 219

StdP: 98.72

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -36.8	(c) -33.7	(d) -39.6	(e) 0.1	(f) -36.6	(g) -36.6	(h) 0.1	(i) -33.6	(j) 14.0	(k) -7.9	(l) 12.2	(m) -7.2	(n) 0.6	(o) 250	(p) 0.617

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.8	30.5	18.4	28.7	17.8	27.1	17.3	20.5	26.6	19.6	25.7	18.7	24.7	3.6	30

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.6	13.8	23.0	17.5	12.9	22.1	16.5	12.1	21.3	60.3	26.6	57.1	25.7	54.2	24.7	24.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 11.4	(b) 9.9	(c) 8.5	(e) -40.0	(f) 33.8	(g) 3.8	(h) 2.0	(i) -42.7	(j) 35.2	(k) -45.0	(l) 36.4	(m) -47.1	(n) 37.6	(o) -49.9	(p) 39.1		
(a) 11.4	(b) 9.9	(c) 8.5	(e) -39.9	(f) 21.9	(g) 3.5	(h) 1.3	(i) -42.4	(j) 22.8	(k) -44.5	(l) 23.6	(m) -46.4	(n) 24.3	(o) -48.9	(p) 25.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	2.2	-17.6	-14.9	-6.8	4.1	12.0	17.8	19.3	17.0	10.6	3.6	-6.2	-13.8
	DBStd	14.40	8.80	7.91	6.83	5.75	5.24	3.95	3.28	3.67	4.46	5.19	8.17	8.25
	HDD10.0	3783	857	697	521	189	41	2	0	1	46	205	487	738
	HDD18.3	6030	1115	931	779	427	203	55	28	69	234	456	737	996
	CDD10.0	932	0	0	0	13	103	237	290	217	64	7	0	0
	CDD18.3	137	0	0	0	0	8	41	59	27	2	0	0	0
	CDH23.3	1622	0	0	0	8	163	463	559	364	64	1	0	0
	CDH26.7	438	0	0	0	1	41	123	158	101	16	0	0	0
(14) Wind	WSAvg	3.2	3.3	3.3	3.5	3.5	3.4	2.7	2.2	2.3	2.6	3.4	3.9	4.0
(15) Precipitation	PrecAvg	407	18	15	20	24	37	51	71	47	32	32	35	26
	PrecMax	578	38	42	48	92	112	124	152	139	76	68	68	57
	PrecMin	234	1	0	4	1	8	15	15	11	4	12	13	10
	PrecStd	94	9	9	12	18	22	27	36	27	18	16	12	11
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	1.6	2.0	9.4	24.3	30.7	31.9	32.7	32.7	28.8	21.3	11.4	2.0
		MCWB	-0.1	0.1	4.5	12.3	16.0	17.8	19.9	19.2	15.1	11.2	6.5	0.3
	2%	DB	-1.0	0.0	5.7	20.2	27.4	29.7	30.3	29.4	24.9	17.7	8.4	0.3
		MCWB	-2.1	-1.4	2.5	11.1	14.7	17.6	19.5	18.2	14.2	9.8	4.8	-0.8
	5%	DB	-3.2	-1.8	3.8	17.1	25.0	27.9	28.3	27.2	21.9	14.7	5.5	-1.4
		MCWB	-4.3	-3.2	1.4	9.4	13.7	17.4	18.9	17.5	13.2	8.3	2.8	-2.6
	10%	DB	-5.4	-3.8	2.0	13.7	22.3	26.1	26.5	25.2	19.4	11.7	3.0	-3.1
		MCWB	-6.3	-5.0	0.2	7.7	12.8	16.8	18.5	16.9	12.1	6.9	1.4	-4.0
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	0.1	0.6	4.8	13.7	17.5	20.6	22.2	21.2	17.1	12.2	7.6	0.8
		MCDB	1.3	1.5	7.7	22.1	27.1	27.3	28.5	27.3	24.5	18.9	9.8	1.4
	2%	WB	-1.9	-1.3	2.9	11.6	15.9	19.5	21.0	19.7	15.2	10.4	4.7	-0.7
		MCDB	-0.9	-0.1	4.9	18.4	24.2	26.4	27.0	26.6	22.3	16.3	7.9	0.3
	5%	WB	-4.2	-3.1	1.6	9.8	14.7	18.6	20.1	18.6	14.0	8.8	2.9	-2.4
		MCDB	-3.2	-2.0	3.4	16.2	22.7	25.3	25.7	25.4	20.8	13.9	5.0	-1.6
	10%	WB	-6.3	-5.0	0.4	8.2	13.5	17.7	19.3	17.6	12.7	7.4	1.3	-4.0
		MCDB	-5.4	-3.9	2.0	13.6	20.9	24.2	24.6	23.8	18.6	11.2	3.0	-3.0
(35) Mean Daily Temperature Range	MDBR		9.4	10.5	10.2	11.0	12.8	11.5	10.8	12.2	12.3	9.6	7.2	8.4
	5% DB	MCDBR	8.2	8.8	10.0	16.9	17.4	15.2	13.9	16.1	17.7	16.2	8.6	8.2
		MCWBR	7.7	7.9	7.5	8.9	7.3	6.0	5.1	6.6	8.3	9.1	6.1	7.7
	5% WB	MCDBR	8.5	8.7	8.7	15.1	14.1	12.1	11.2	13.6	15.2	14.6	7.7	8.2
		MCWBR	8.1	8.0	6.7	8.6	6.7	5.4	5.1	6.0	7.7	8.7	5.6	8.0
(40) Clear-Sky Solar Irradiance	taub		0.296	0.332	0.384	0.410	0.411	0.408	0.431	0.409	0.371	0.355	0.338	0.303
	taud		2.022	1.958	1.905	2.117	2.192	2.284	2.215	2.254	2.305	2.232	2.050	1.967
	Ebn at Noon		635	729	770	815	838	844	814	811	797	713	571	528
	Edh at Noon		78	114	150	139	135	125	132	120	101	87	76	69
(44) All-Sky Solar Radiation	RadAvg		1.13	2.20	3.61	4.67	5.68	6.28	5.67	4.88	3.49	1.83	0.93	0.77
	RadStd		0.09	0.15	0.34	0.29	0.51	0.44	0.34	0.32	0.33	0.25	0.14	0.11

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (11 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page